Computer Laboratory Report - Medical Imaging Simulations: Radiation

Introduction

This computer laboratory evaluates the numerical approximation of the integral

$$\int_0^{7,5} \frac{x^{x+1}}{\ln(x+3)\, e^x} dx \tag{1}$$

using the Monte Carlo (MC) method—a stochastic numerical technique—by employing “rectangles” and "darts" integration, which rely on sampling random points, corresponding to the number of rectangles and darts, respectively. The purpose of the laboratory is to approximate the integral numerically using random sampling on the definite interval and study how accuracy improves with the number of samples. The integrand in (1) is given as:

$$f(x) = \frac{x^{x+1}}{\ln(x+3)\, e^x} \tag{2}$$

Method

For both methods, simulations were performed using varying numbers of samples:

$$N = [10, 100, 1000, 10000]$$

Integrals of both methods were the approximated using standard Monte Carlo integration formula, with upper and lower boundaries as $a = 0$ and $b = 7.5$, respectively. For each sample size, the integration was repeated over 20 independent runs to compute the mean estimate and statistical variance. To visually assess the performance of both MC integration techniques, several plots were generated for each method. These plots help analyze the evolution of the integral estimate and its variance.

Rectangles method

The Rectangles method is based on uniform random sampling of points along the interval. These points are then used to compute the function values, i.e., the rectangle height, to approximate the area under the curve as the mean of sampled function values multiplied by the interval width, $b - a$, (3).

$$\text{Integral} \approx \frac{\text{sum of all rectangle areas}}{\text{number of rectangles}} = \frac{\sum((b-a)\cdot f(x_i)}{\text{number of rectangles}} = \frac{(b-a)}{N}\sum_{i=1}^{N} f(x_i) \qquad (3)$$

, where N is the number of randomized samples, $i = 1, 2, \dots, N$.

Darts Method

The Darts Method estimates the value of the definite integral by simulating uniform random points, i.e., *darts*, thrown inside a bounding rectangle that covers the graph of the function (2). Each dart is assigned coordinates $\big(x_i, f(x_i)\big)$, with $x_i \in [a, b]$ and $f(x_i) \in [0, y_{max}]$, where y_{max} is the maximum of $f(x)$ within the interval. A *hit* occurs when the randomly generated y_i value lies below the curve, i.e., satisfying $y_i \leq f(x_i)$; otherwise, it considered a *miss*.

In the Darts Method, the area under the curve is approximated by calculating the ratio of darts that fall below the curve. This fraction represents the proportion of the area under the function relative to the area of the bounding rectangle.

$$\frac{\text{Integral}}{A_{\text{box}}} \approx \frac{\text{Number of darts under curve}}{\text{Total darts}} \qquad (4)$$

Thus, this gives the value of the definite integral as:

$$\text{Integral} \approx \frac{\text{hits}}{\text{throws}} \cdot (b-a) \cdot y_{\max} \qquad (5)$$

Results

Results of the Rectangles Method

Table 1 presents the numerical results obtained using the Rectangles Method. The table displays the mean estimates of the integral obtained from 20 runs for each sample size, along with the corresponding variance. As the sample size increases, the estimates become more accurate, and the variance decreases.

Table 1. Rectangles Method Integral and Variance Estimates

Number of Rectangles (N)	Average Integral Estimate	Variance
10	2798.9	$6.06 \cdot 10^6$
100	3669.0	$1.60 \cdot 10^6$
1000	3207.3	$7.93 \cdot 10^4$
10000	3153.1	$2.84 \cdot 10^3$

Figure 1 presents two logarithmic plots illustrating the evolution of the integral estimation and its variance as a function of the number of rectangles N (or samples). The top plot features the estimated value and the exact value of the integral. The bottom subplot displays the variance of the integral estimation.

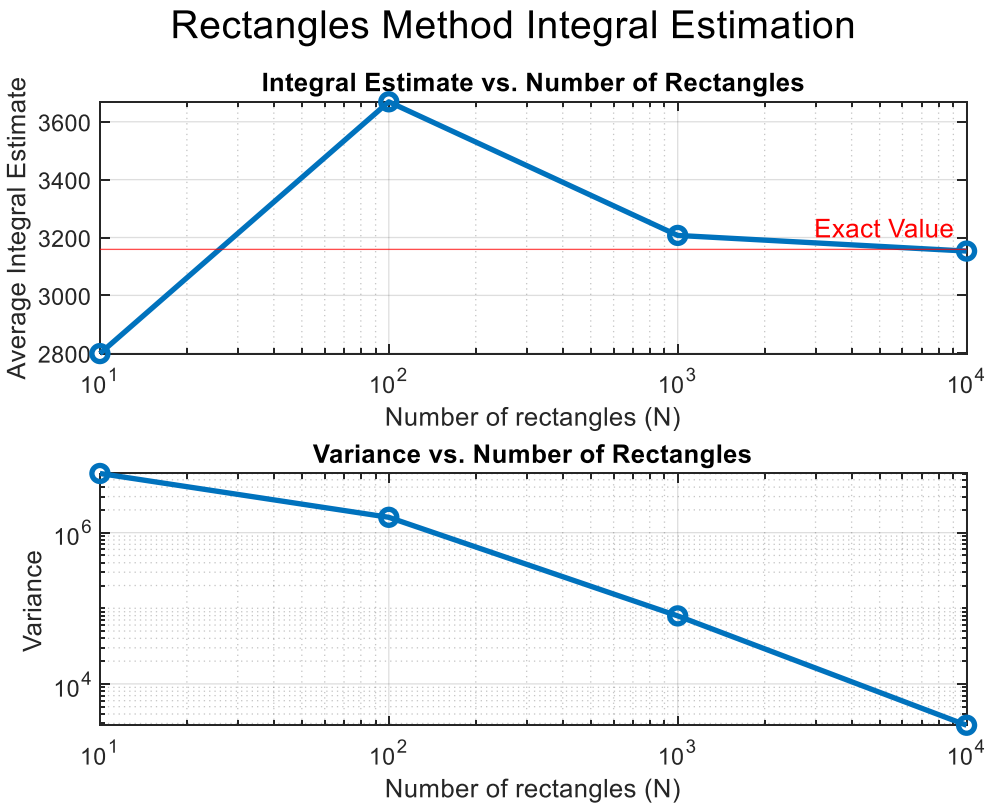


Figure 1: Logarithmic plots illustrating the evolution of the integral estimates as a function of N and the evolution of the variance as a function of N.

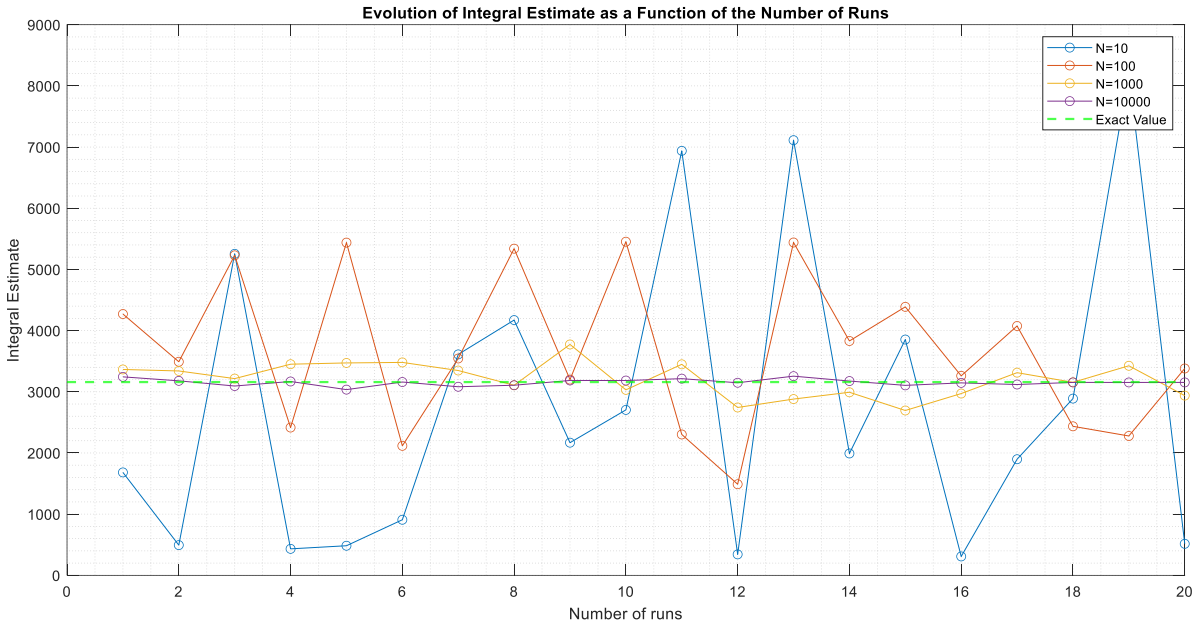


Figure 2: Integral estimates over 20 runs per N to visualize the fluctuations across the runs for each number of samples.

Results of the Darts Method

Table 2 presents the results of the average integral estimates along with their corresponding variances from 20 runs. The key observation is that no discernible pattern emerges among the average estimates; however, increasing the sample size leads to a more accurate approximation. Additionally, a consistent decrease in variance is evident.

Table 2. Rectangles Method Integral and Variance Estimates.

Number of Darts (N)	Average Integral Estimate	Variance
10	1934.7	$5.91 \cdot 10^6$
100	3216.4	$1.68 \cdot 10^6$
1000	3117.3	$1.68 \cdot 10^5$
10000	3187.6	$21.15 \cdot 10^3$

Figure 3 is present of two logarithmic plots. The top subplot illustrates how the evolution of the mean integral estimate improves with increasing N. At low sample sizes, the estimates deviate significantly from the exact value, but as N increases, the estimates begin to converge and stabilize near the exact solution, marked by a dashed green line. The bottom subplot depicts the variance of the integral estimates as a function of the number of darts.

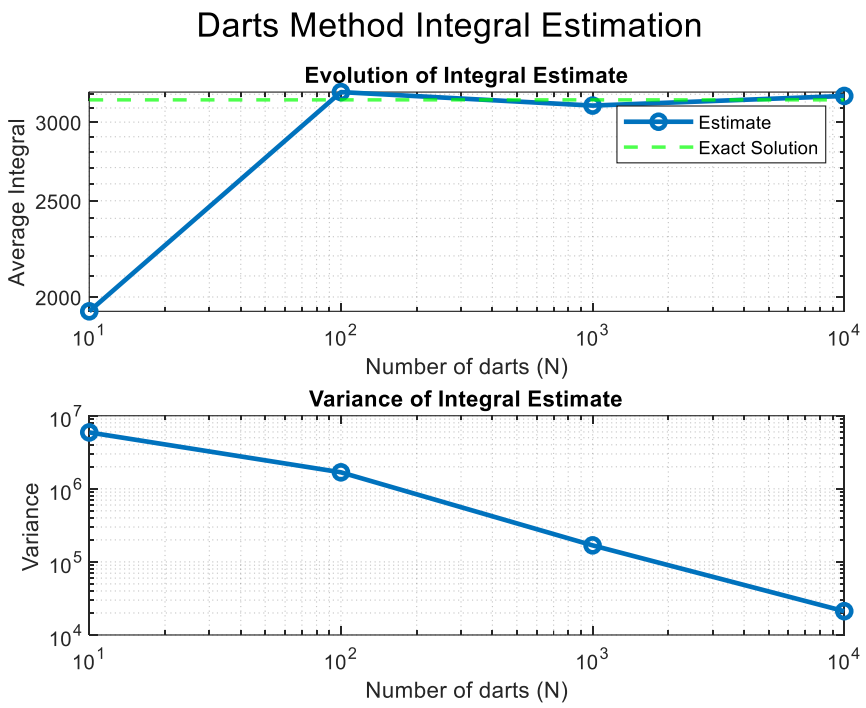


Figure 3: Logarithmic plots illustrating the evolution of the integral estimates as a function of N and the evolution of the variance as a function of N.

Figure 4 displays the integral estimates from 20 independent runs for each value of N. For lower samples sizes, highly erratic results are produced, with significant fluctuations between runs. Conversely, higher sample sizes lead to more accurate results around the true value, indicating increased stability and reduced randomness.

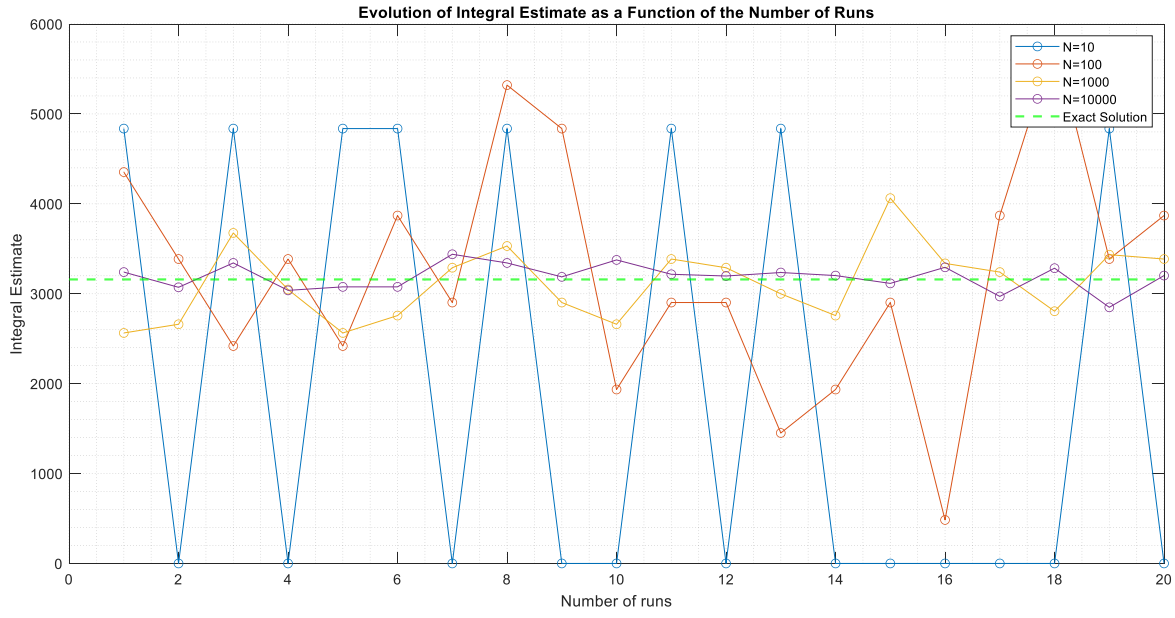


Figure 4: Darts method integral estimates over 20 runs per N to visualize the fluctuations across the runs for each number of samples.

Discussion

Since the integral is estimated using random points, the results will vary each time the program is executed. The following discussion pertains to the results above, and different conclusions may be reached based on the outcomes of various executions.

Discussion of Rectangles Method

For $N = 10$, the mean estimate is relatively low (2798.9) and the variance very high ($6.06 \cdot 10^6$). For small samples, the variance is significantly high, reflecting the large fluctuations in the computed estimates due to limited sampling. The mean estimate for $N = 100$ is over the exact value however, the variance decreases to $1.60 \cdot 10^6$. For $N = 1000$, the mean estimate is very close to the exact values, and the variance decreases under 80000. For $N = 10000$, the estimate is more accurate, with a variance around 2800, indicating convergence to a more accurate value.

The results show that the accuracy of the Rectangles Method depends on the number of samples N; as N increases, the mean integral value approaches the exact value. The second observation is the variance, which decreases significantly with a larger number of samples.

Discussion of Darts Method

For a small number of samples (N), the darts method yields an estimated integral that is significantly lower than the exact value, resulting in high variance across different runs. However, when $N = 100$, the average estimate improves greatly and becomes more accurate, and the variance decreases to $1.68 \cdot 10^6$. For larger sample sizes, $N = 1000$ and $N = 10000$, the mean estimates are more accurate, and a steady decrease in variance is noticeable, as shown in Table 2.

The results support the theory of the law of large numbers, which states that larger sample sizes lead to improved accuracy. The Darts Method can be particularly beneficial for functions that have irregular shapes and complexities, especially when dealing with integrands that are challenging to compute analytically. Additionally, because the Darts Method relies heavily on the geometry of the bounding box, exploring different geometrical shapes for the bounding box can significantly enhance the accuracy of the estimates.

Conclusion

For larger samples, the variance decreases, following the expected behavior of the MC method, since, as N increases, the Law of Large Numbers ensures that the mean of the samples converges to the true integral value while the variance diminishes. This behavior is evident in the figure, which shows that increasing the number of samples improves both accuracy and stability.

The darts method is an intuitive approach to estimating integrals, but it is less efficient in terms of variance compared to the rectangles method. While the darts method is relatively simple, it may require some optimization for practical applications.

While the method is geometrically intuitive, its accuracy is heavily dependent on the bounding box. Furthermore, large variability can arise, especially when the actual area under the curve is small relative to the total area of the bounding box. In contrast to the Rectangles Method, which samples function values along a one-dimensional domain, the Darts Method samples random points in a two-dimensional region, considering the hit or miss criterion for each point.

MATLAB code

```
%% Computer Laboratory: Monte Carlo Integration

% Evaluate the following integral using the "darts" and the "rectangles" method.
% Study the evolution of your estimate and its variance as a function of number of
samples.
% Write a short report explaining your codes and with your results presented in
graphs.

clc; clear; close all;


% === PARAMETERS ===


% Define anonymous (integrand) function
f = @(x) (x.^(x + 1)) ./ (log(x + 3) .* exp(x));

% Integration bounds
a = 0;
b = 7.5;

% Exact value of the integral (for comparison)
INTEGRAL = integral(f, a, b, 'RelTol', 1e-12, 'AbsTol', 1e-12);

% Sampling settings
N_vec = [1e1, 1e2, 1e3, 1e4];       % number of rectangles/darts
RUNS = 20;                          % number of runs



% === RECTANGLES METHOD ===

tic;
% Preallocate storage
rec_integral_mat = zeros(RUNS, length(N_vec));      % integral matrix
rec_integral_vec = zeros(1,    length(N_vec));      % average integral vector (per
N)
rec_variance_vec = zeros(1,    length(N_vec));      % variance vector (per N)

% Monte Carlo Integration
for n = 1:length(N_vec)

    N = N_vec(n);

    for run = 1:RUNS

        % Sample N random x-values in [a, b]
        randX_vec = a + (b-a) * rand([N,1]);
        randY_vec = f(randX_vec);

        % Estimate integral as mean of all areas = width x height (per run)
        rec_integral_mat(run,n) = mean((b-a) * randY_vec);

    end

    % Compute mean integral and variance (per N)
```

```
    rec_integral_vec(n) = mean(rec_integral_mat(:,n));
    rec_variance_vec(n) = var(rec_integral_mat(:,n));

end

% Table
T_rec = table(N_vec', rec_integral_vec', rec_variance_vec', ...
    'VariableNames', {'N','Avergage Integral' , 'Variance'});
disp(T_rec)

% Plot: Integral estimates across runs
figure;
plot(1:RUNS, rec_integral_mat, "-o");
yline(INTEGRAL,"--", "Color", "g", "LineWidth", 1.5)
legend("N=10", "N=100", "N=1000", "N=10000", "Exact Solution");
xlabel('Number of Runs');
ylabel('Integral Estimate');
title('Evolution of Rectangles Method Integral Estimate');
grid minor;

% Plot: Average and Variance vs N
figure;
sgtitle('Rectangles Method: Convergence Analysis');

% Top plot: evolution of the integral estimate
subplot(2,1,1);
loglog(N_vec, rec_integral_vec, '-o','LineWidth',2);
yline(INTEGRAL, "--", "Color", "g", "LineWidth", 1.5)
grid minor;
legend("Estimate", "Exact Solution");
xlabel('Number of Rectangles (N)');
ylabel('Average Estimate');
title('Average Integral Estimate');

% Bottom plot: variance of the integral estimate
subplot(2,1,2);
loglog(N_vec, rec_variance_vec, '-o','LineWidth',2);
grid minor;
xlabel('Number of Rectangles (N)');
ylabel('Variance');
title('Variance of Integral Estimate');

time_end_1 = toc;

% === DARTS METHOD ===

% Preallocate storage
dar_integral_mat = zeros(RUNS, length(N_vec));      % integral matrix
dar_integral_vec = zeros(1,    length(N_vec));      % average integral vector (per
N)
dar_variance_vec = zeros(1,    length(N_vec));      % variance vector (per N)


% Monte Carlo Integration
tic;
for n = 1:length(N_vec)

    N = N_vec(n);
```

```
    % Define interval limit
    x_interval = linspace(a, b, N)';
    f_interval = f(x_interval);

    % Find maximum value of f(x) in the interval [a,b]
    y_max = max(f_interval);

    for run = 1:RUNS

        % Generate N random x-values uniformly in [a, b]
        randX_vec = a + (b-a) * rand([N,1]);

        % Generate N random y-values uniformly in [0, y_max]
        randY_vec = y_max * rand([N,1]);

        % Evaluate function at those random x points
        randF_vec = f(randX_vec);

        % Logical vector: true where random y falls below the dart (x,f(x))
        inside_vec = randY_vec <= randF_vec;

        % Count hits (darts below curve) for N throws
        throws = N;
        hits = sum(inside_vec);

        % Estimate area using hit ratio
        area_box = (b-a) * y_max;
        dar_integral_mat(run,n) = (hits / throws) * area_box;

    end

    % Average per N
    dar_integral_vec(n) = mean(dar_integral_mat(:,n));
    dar_variance_vec(n) = var(dar_integral_mat(:,n));

end
time_end_2 = toc;

% === DISPLAY ===

% TABLE

T_dar = table(N_vec', dar_integral_vec', dar_variance_vec', 'VariableNames',
{'N','Avergage Integral' , 'Variance'});
disp(T_dar)

% PLOTS

% Integral vs. Number of Runs
figure;
plot(1:RUNS, dar_integral_mat(:,:), "-o");
yline(INTEGRAL, "--", "Color", "g", "LineWidth", 1.5)
grid minor;
legend("N=10", "N=100", "N=1000", "N=10000", "Exact Solution");
xlabel('Number of runs');
ylabel('Integral Estimate');
title('Evolution of Darts Method Integral Estimate');

% Average Integral & Variance vs. Number of Darts
```

```
figure;
sgtitle('Darts Method Integral Estimation');

% Top plot: evolution of the integral estimate
subplot(2,1,1);
loglog(N_vec, dar_integral_vec, '-o','LineWidth',2);
yline(INTEGRAL, "--", "Color", "g", "LineWidth", 1.5)
grid minor;
legend("Estimate", "Exact Solution");
xlabel('Number of darts (N)');
ylabel('Average Integral');
title('Evolution of Integral Estimate');

% Bottom plot: variance of the integral estimate
subplot(2,1,2);
loglog(N_vec, dar_variance_vec, '-o','LineWidth',2);
grid minor;
xlabel('Number of darts (N)');
ylabel('Variance');
title('Variance of Integral Estimate');



% === METHOD DIFFERENCE ===


% Absolute Integral Difference
integral_diff_vec = abs(rec_integral_vec - dar_integral_vec);

% Absolute Variance Difference
variance_diff_vec = abs(rec_variance_vec - dar_variance_vec);


% Display table of differences
T_diff = table(N_vec', integral_diff_vec', variance_diff_vec', ...
    'VariableNames', {'N', 'Integral_Difference', 'Variance_Difference'});
disp(T_diff)


% Plot comparison of method differences
figure;
sgtitle('Integration Method Comparison');

% Top plot: integral estimate difference
subplot(2,1,1);
loglog(N_vec, integral_diff_vec, '-o','LineWidth',2);
grid minor;
xlabel('Number of Darts (N)');
ylabel('|Average Integral|');
title('Absolute Difference in Integral Estimates');

% Bottom plot: evolution of variance difference
subplot(2,1,2);
loglog(N_vec, variance_diff_vec, '-o','LineWidth',2);
grid minor;
xlabel('Number of Darts (N)');
ylabel('|Variance|');
title('Absolute Difference in Variance');
```

```
error_abs = zeros(2,4);
error_abs(1,:) = abs(rec_integral_vec - INTEGRAL) / abs(INTEGRAL);
error_abs(2,:) = abs(dar_integral_vec - INTEGRAL) / abs(INTEGRAL);

error_rel = zeros(2,4);
error_rel(1,:) = abs(rec_integral_vec - INTEGRAL) * 100 / abs(rec_integral_vec);
error_rel(2,:) = abs(dar_integral_vec - INTEGRAL) * 100 / abs(dar_integral_vec);
```